Class 9 Mathematics — Exercise 4.4

Basic Concept

Key Words

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HCF	lowest power and common factors
LCM	highest power and every factor

- **LCM** of two or more polynomials is the polynomial of lowest degree that is exactly divisible by each of the given polynomials.
- **LCM = Common factors × Uncommon factors**

HCF by Division Method

1. Write both polynomials in descending powers. Take the **higher degree** polynomial as **dividend** and the other as **divisor**.
2. Divide. If needed, **multiply the dividend by a number** to avoid fractions.
3. A **common numerical factor** of a remainder can be **ignored** (it does not affect the HCF).
4. The remainder becomes the new **divisor** and the old divisor becomes the new **dividend**.
5. Continue until the remainder is **0**. The **last divisor** is the HCF.

Question 1

Give quick answers without doing any procedure. If HCF of $x^3 + 5x^2 + 6x$ and $x^3 + 9x^2 + 14x$ is $x^2 + 2x$, then:

(i) HCF of $5(x^3 + 5x^2 + 6x)$ and $x^3 + 9x^2 + 14x$ $= \boxed{x^2 + 2x}$

(ii) HCF of $x^3 + 5x^2 + 6x$ and $2(x^3 + 9x^2 + 14x)$ $= \boxed{x^2 + 2x}$

(iii) HCF of $3(x^3 + 5x^2 + 6x)$ and $7(x^3 + 9x^2 + 14x)$ $= \boxed{x^2 + 2x}$

(iv) HCF of $15(x^3+5x^2+6x)$ and $25(x^3+9x^2+14x)$ $= \boxed{5(x^2 + 2x)}$ (because 5 is common in 15 and 25)

(v) Does the HCF remain unchanged if both are multiplied by x?

$$x(x^3 + 5x^2 + 6x), \qquad x(x^3 + 9x^2 + 14x)$$

No, the HCF becomes $\boxed{x(x^2 + 2x)}$.

Question 2

Find the LCM of the following monomials by completing the table.

Monomials	LCM of coeff.	LCM of x	LCM of y	LCM of z	LCM
(i) $8x^6y,\ 4x^2yz$	8	x^6	y	z	$8x^6yz$
(ii) $12x^2y^4z,\ 24x^3z$	24	x^3	y^4	z	$24x^3y^4z$
(iii) $18x^3z,\ 9xy^2z,\ 6x^6yz^3$	18	x^6	y^2	z^3	$18x^6y^2z^3$
(iv) $xyz^3,\ z^5y^3x,\ 28x^3y^5z$	28	x^3	y^5	z^5	$28x^3y^5z^5$

LCM of coefficients:

$$\begin{array}{r|l} 2 & 8,\ 4 \\ 2 & 4,\ 2 \\ 2 & 2,\ 1 \\ \hline & 1,\ 1 \end{array} = 8 \qquad \begin{array}{r|l} 2 & 12,\ 24 \\ 2 & 6,\ 12 \\ 2 & 3,\ 6 \\ 3 & 3,\ 3 \\ \hline & 1,\ 1 \end{array} = 24 \qquad \begin{array}{r|l} 2 & 18,\ 9,\ 6 \\ 3 & 9,\ 9,\ 3 \\ 3 & 3,\ 3,\ 1 \\ \hline & 1,\ 1,\ 1 \end{array} = 18 \qquad 1, 1, 28 \Rightarrow 28$$

Question 3

Find the HCF by division method: $a^2 + a - 2,\quad a^3 + 2a^2 + a + 2$

Step 1: Divide $a^3 + 2a^2 + a + 2$ by $a^2 + a - 2$:

$$a^3 + 2a^2 + a + 2 = (a^2 + a - 2)(a + 1) + (2a + 4)$$

Remainder $= 2a + 4 = 2(a + 2)$. Ignore 2 $\Rightarrow$ new divisor $= a + 2$

Step 2: Divide $a^2 + a - 2$ by $a + 2$:

$$a^2 + a - 2 = (a + 2)(a - 1) + 0$$

Remainder $= 0$, so the last divisor is the HCF.

$$\text{H.C.F} = \boxed{a + 2} \text{ Ans}$$

Question 4

$$x^3 + 2x^2 - 4x - 8,\quad 2x^3 + 7x^2 + 4x - 4$$

Step 1: Divide $2x^3 + 7x^2 + 4x - 4$ by $x^3 + 2x^2 - 4x - 8$:

$$2x^3 + 7x^2 + 4x - 4 = (x^3 + 2x^2 - 4x - 8)(2) + (3x^2 + 12x + 12)$$

Remainder $= 3x^2 + 12x + 12 = 3(x^2 + 4x + 4)$. Ignore 3 $\Rightarrow$ new divisor $= x^2 + 4x + 4$

Step 2: Divide $x^3 + 2x^2 - 4x - 8$ by $x^2 + 4x + 4$:

$$x^3 + 2x^2 - 4x - 8 - x(x^2 + 4x + 4) = -2x^2 - 8x - 8 = -2(x^2 + 4x + 4)$$

Ignore -2; the remaining part is exactly $x^2 + 4x + 4$, so the division leaves remainder 0.

$$\text{H.C.F} = \boxed{x^2 + 4x + 4 = (x+2)^2} \text{ Ans}$$

Question 5

$$2x^3 + x^2 - x - 2, \quad 3x^3 - x^2 + x - 3$$

Step 1: Multiply $3x^3 - x^2 + x - 3$ by 2 (to avoid fractions) and divide by $2x^3 + x^2 - x - 2$:

$$6x^3 - 2x^2 + 2x - 6 - 3(2x^3 + x^2 - x - 2) = -5x^2 + 5x = -5(x^2 - x)$$

Ignore $-5 \Rightarrow$ new divisor $= x^2 - x$

Step 2: Divide $2x^3 + x^2 - x - 2$ by $x^2 - x$:

$$2x^3 + x^2 - x - 2 = (x^2 - x)(2x + 3) + (2x - 2)$$

Remainder $= 2x - 2 = 2(x - 1)$. Ignore $2 \Rightarrow$ new divisor $= x - 1$

Step 3: Divide $x^2 - x$ by $x - 1$:

$$x^2 - x = (x - 1)(x) + 0$$

$$\text{H.C.F} = \boxed{x - 1} \text{ Ans}$$

Question 6

$$3 + 2p^4 + 5p^2, \quad 5p + 5p^3 + 3 + 3p^2$$

In descending order: $2p^4 + 5p^2 + 3, \quad 5p^3 + 3p^2 + 5p + 3$

Step 1: Multiply $2p^4 + 5p^2 + 3$ by 5 (to avoid fractions) and divide by $5p^3 + 3p^2 + 5p + 3$:

$$10p^4 + 25p^2 + 15 - 2p(5p^3 + 3p^2 + 5p + 3) = -6p^3 + 15p^2 - 6p + 15$$

$$= -3(2p^3 - 5p^2 + 2p - 5) \qquad \text{Ignore } -3$$

Multiply $2p^3 - 5p^2 + 2p - 5$ by 5 and continue the division:

$$10p^3 - 25p^2 + 10p - 25 - 2(5p^3 + 3p^2 + 5p + 3) = -31p^2 - 31$$

Remainder $= -31p^2 - 31 = -31(p^2 + 1)$. Ignore $-31 \Rightarrow$ new divisor $= p^2 + 1$

Step 2: Divide $5p^3 + 3p^2 + 5p + 3$ by $p^2 + 1$:

$$5p^3 + 3p^2 + 5p + 3 = (p^2 + 1)(5p + 3) + 0$$

$$\text{H.C.F} = \boxed{p^2 + 1} \text{ Ans}$$

Question 7

$$24x^4 - 2x^3 - 60x^2 - 32x, \quad 18x^4 - 6x^3 - 39x^2 - 18x$$

$$= 2(12x^4 - x^3 - 30x^2 - 16x), \qquad 3(6x^4 - 2x^3 - 13x^2 - 6x)$$

The numbers 2 and 3 have no common factor, so we find the HCF of the brackets.

Step 1: Divide $12x^4 - x^3 - 30x^2 - 16x$ by $6x^4 - 2x^3 - 13x^2 - 6x$:

$$12x^4 - x^3 - 30x^2 - 16x = (6x^4 - 2x^3 - 13x^2 - 6x)(2) + (3x^3 - 4x^2 - 4x)$$

Step 2: Divide $6x^4 - 2x^3 - 13x^2 - 6x$ by $3x^3 - 4x^2 - 4x$:

$$6x^4 - 2x^3 - 13x^2 - 6x = (3x^3 - 4x^2 - 4x)(2x + 2) + (3x^2 + 2x)$$

Step 3: Divide $3x^3 - 4x^2 - 4x$ by $3x^2 + 2x$:

$$3x^3 - 4x^2 - 4x = (3x^2 + 2x)(x) - 6x^2 - 4x = (3x^2 + 2x)(x - 2) + 0$$

$$\text{H.C.F} = 3x^2 + 2x = \boxed{x(3x + 2)} \text{ Ans}$$

Question 8

$$2x^3 + 6x^2 + x + 3, \quad 3x^3 + 9x^2 - 2x - 6, \quad x^3 + 3x^2 + 2x + 6$$

First find the HCF of the last two, then the HCF of that answer with the first.

Step 1: Divide $3x^3 + 9x^2 - 2x - 6$ by $x^3 + 3x^2 + 2x + 6$:

$$3x^3 + 9x^2 - 2x - 6 = (x^3 + 3x^2 + 2x + 6)(3) + (-8x - 24)$$

Remainder $= -8x - 24 = -8(x + 3)$. Ignore $-8 \Rightarrow$ new divisor $= x + 3$

Step 2: Divide $x^3 + 3x^2 + 2x + 6$ by $x + 3$:

$$x^3 + 3x^2 + 2x + 6 = (x + 3)(x^2 + 2) + 0 \quad \Rightarrow \quad \text{HCF of these two} = x + 3$$

Step 3: Divide $2x^3 + 6x^2 + x + 3$ by $x + 3$:

$$2x^3 + 6x^2 + x + 3 = (x + 3)(2x^2 + 1) + 0$$

$$\text{H.C.F} = \boxed{x + 3} \text{ Ans}$$

Question 9

Find the LCM: $9a^2b - b, \quad 6a^2 + 2a$

$9a^2b - b$	$6a^2 + 2a$
$= b(9a^2 - 1)$	$= 2a(3a + 1)$

$9a^2b - b$	$6a^2 + 2a$
$= b[(3a)^2 - (1)^2]$	
$= b(3a+1)(3a-1)$	

Common factor $= (3a+1)$; Uncommon factors $= b,\ (3a-1),\ 2a$

$$\begin{aligned}\text{L.C.M} &= \text{Common factors} \times \text{Uncommon factors}\\ &= (3a+1) \times b(3a-1)(2a)\\ &= 2ab(3a+1)(3a-1)\\ &= \boxed{2ab(9a^2-1)} \text{ Ans}\end{aligned}$$

Question 10

$$p^3q - pq^3, \quad p^5q^2 - p^2q^5$$

$p^3q - pq^3$	$p^5q^2 - p^2q^5$
$= pq(p^2 - q^2)$	$= p^2q^2(p^3 - q^3)$
$= pq(p+q)(p-q)$	$= (pq)^2(p-q)(p^2+pq+q^2)$
	$= (pq)(pq)(p-q)(p^2+pq+q^2)$

$$\begin{aligned}\text{L.C.M} &= \text{Common factors} \times \text{Uncommon factors}\\ &= (pq)(p-q) \times (p+q)(pq)(p^2+pq+q^2)\\ &= p^2q^2(p+q)(p-q)(p^2+pq+q^2)\\ &= \boxed{p^2q^2(p+q)(p^3-q^3)} \text{ Ans}\end{aligned}$$

Question 11

$$4x^2y - y, \quad 2x^2 + x$$

$4x^2y - y$	$2x^2 + x$
$= y(4x^2 - 1)$	$= x(2x+1)$
$= y[(2x)^2 - (1)^2]$	
$= y(2x+1)(2x-1)$	

$$\begin{aligned}\text{L.C.M} &= \text{Common factors} \times \text{Uncommon factors}\\ &= (2x+1) \times y(2x-1)(x)\\ &= \boxed{xy(2x+1)(2x-1)} \text{ Ans}\end{aligned}$$

Question 12

$$x^2 - x - 6, \quad x^2 + x - 2, \quad x^2 - 4x + 3$$

$x^2 - x - 6$	$x^2 + x - 2$	$x^2 - 4x + 3$
$(-6 \to -3, 2)$	$(-2 \to 2, -1)$	$(3 \to -3, -1)$
$= x^2 - 3x + 2x - 6$	$= x^2 + 2x - x - 2$	$= x^2 - 3x - x + 3$
$= x(x-3) + 2(x-3)$	$= x(x+2) - 1(x+2)$	$= x(x-3) - 1(x-3)$
$= (x+2)(x-3)$	$= (x+2)(x-1)$	$= (x-3)(x-1)$

$$\text{L.C.M} = \boxed{(x-3)(x-1)(x+2)} \text{ Ans}$$

Question 13

$$m^6 - 1, \quad m^4 - 1, \quad m^3 - 1$$

(i)

$$m^6 - 1 = (m^3)^2 - (1)^2 = (m^3+1)(m^3-1)$$
$$= [(m)^3 + (1)^3][(m)^3 - (1)^3]$$
$$= (m+1)[(m)^2 - (m)(1) + (1)^2]\ (m-1)[(m)^2 + (m)(1) + (1)^2]$$
$$= (m+1)(m^2 - m + 1)(m-1)(m^2 + m + 1)$$

(ii)

$$m^4 - 1 = (m^2)^2 - (1)^2 = (m^2+1)(m^2-1)$$
$$= (m^2+1)[(m)^2 - (1)^2] = (m^2+1)(m+1)(m-1)$$

(iii)

$$m^3 - 1 = (m-1)(m^2 + m + 1)$$

$$\text{L.C.M} = \text{Common factors} \times \text{Uncommon factors}$$
$$= (m-1) \times (m+1)(m^2 - m + 1)(m^2 + m + 1)(m^2 + 1)$$
$$= (m-1)(m^2 + m + 1)(m+1)(m^2 - m + 1)(m^2 + 1)$$
$$= \boxed{(m^3-1)(m^3+1)(m^2+1) = (m^6-1)(m^2+1)} \text{ Ans}$$

Question 14

$$x^3 + 2x^2 - x - 2, \quad x^2 - x - 2, \quad x^2 - 4$$

x^3+2x^2-x-2	x^2-x-2	x^2-4
$=x^2(x+2)-1(x+2)$	$=x^2-2x+x-2$	$=x^2-2^2$
$=(x+2)(x^2-1)$	$=x(x-2)+1(x-2)$	$=(x+2)(x-2)$
$=(x+2)(x+1)(x-1)$	$=(x-2)(x+1)$	

$$\text{L.C.M} = \text{Common factors} \times \text{Uncommon factors}$$
$$= (x+2)(x+1)(x-1)(x-2)$$
$$= (x+2)(x-2)(x+1)(x-1)$$
$$= (x^2-2^2)(x^2-1^2)$$
$$= (x^2-4)(x^2-1)$$
$$= x^2(x^2-1)-4(x^2-1)$$
$$= x^4-x^2-4x^2+4$$
$$= \boxed{x^4-5x^2+4} \text{ Ans}$$

Question 15

$$x^2+x-20, \quad x^2-10x+24, \quad x^2-x-30$$

x^2+x-20	$x^2-10x+24$	x^2-x-30
$=x^2+5x-4x-20$	$=x^2-6x-4x+24$	$=x^2-6x+5x-30$
$=x(x+5)-4(x+5)$	$=x(x-6)-4(x-6)$	$=x(x-6)+5(x-6)$
$=(x-4)(x+5)$	$=(x-4)(x-6)$	$=(x+5)(x-6)$

$$\text{L.C.M} = \text{Common factors} \times \text{Uncommon factors}$$
$$= \boxed{(x-4)(x+5)(x-6)} \text{ Ans}$$